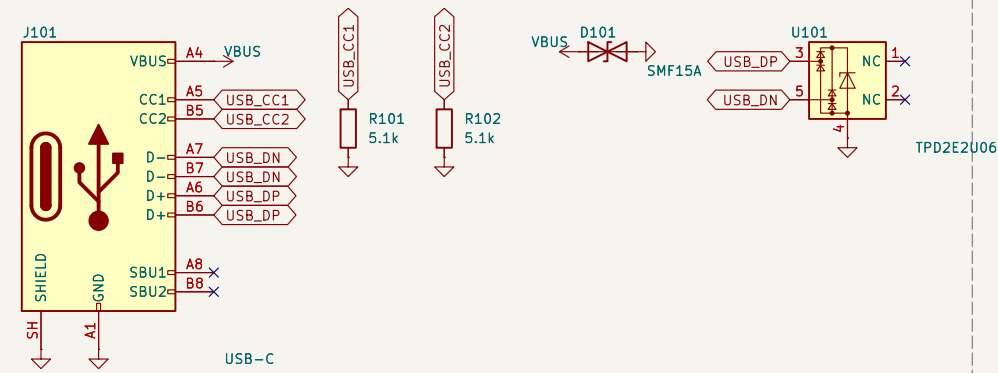




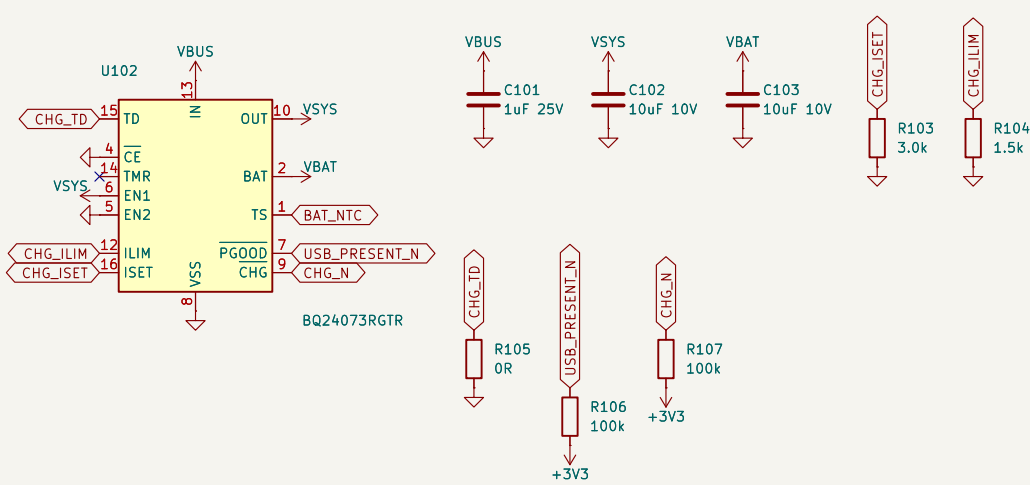
POWER

USB-C ENTRY



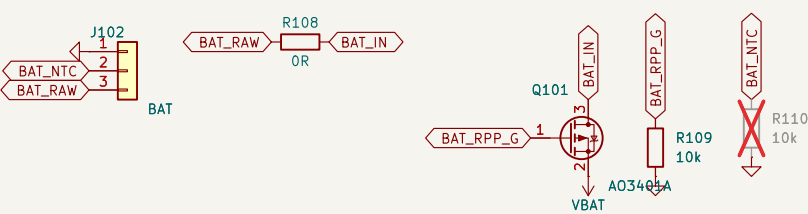
Sink only: 5.1k Rd on CC1/CC2 (ODD JOBS 25). ESD at the connector (119): TPD2E2U06 on D+/D-, SMF15A on VBUS. Shell to GND (plastic enclosure, 26).

CHARGER



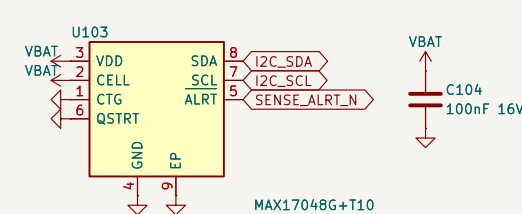
BQ24073 linear charger with power path (DPPM): runs MAO from USB while charging. USB500 Input limit (EN2=0, EN1=1). ISET 3.0k = 297 mA charge. ILIM 1.5k, default timers, pack NTC on TS (0-50 C window).

BATTERY



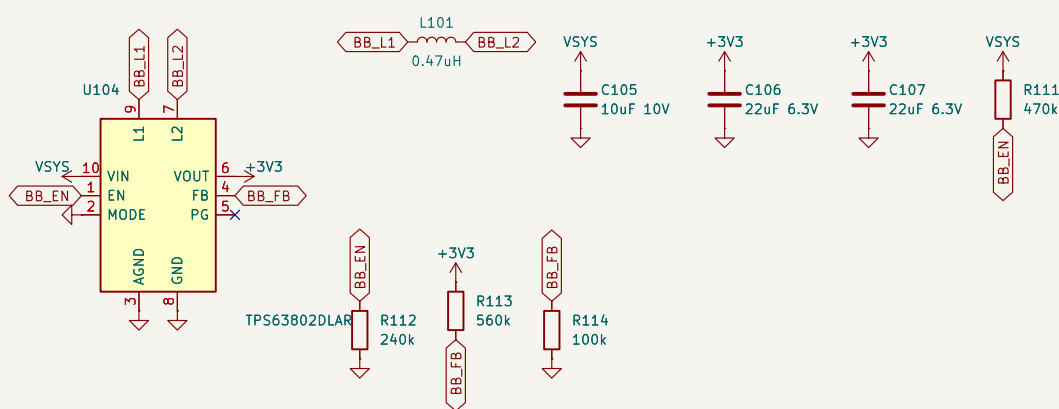
JST SH 3-pin: 1 = BAT-, 2 = NTC, 3 = BAT+ (same order as KINO J1100, ODD JOBS 41/167). Cell MUST carry its own protection PCM and a 10k NTC. R10 = OR link to measure battery current (110). Q1 blocks a reversed cell (51). R11 fits only for cells without NTC.

FUEL GAUGE



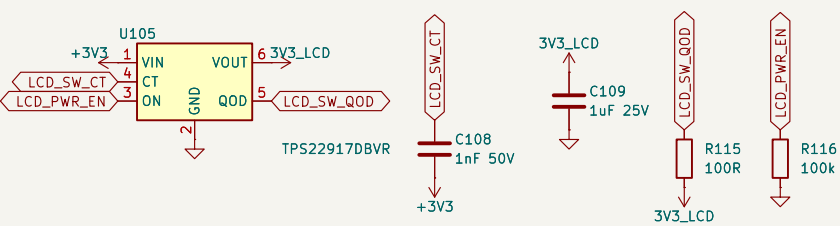
MAX17048 ModelGauge on VBAT, I2C 0x36, ALRT wired-OR with the light sensor INT.

3V3 BUCK-BOOST



TPS63802: VSYS 2.9–4.4 V to +3V3, 2 A, 11 uA Iq, EN divider = hardware UVLO: on at 3.25 V, off at 2.96 V (protects the cell even if firmware fails).
MODE low = PFM, FB 560k/100k = 3.30 V, L = 0.47 uH only.

DISPLAY RAIL



TPS22917 switches 3V3_LCD (panel + backlight). ON pulled down: off until firmware enables it. CT 1 nF: ~4 ms soft start, so panel inrush cannot brown out the MCU. QOD to VOUT: clean power-down.

RAIL SOURCE DECLARATIONS

USB-C entry, charger with power path, battery, fuel gauge, 3.3 V buck-boost, display rail

Sheet: /Power/
File: power.kicad_sch

Title: MAO_MAIN A0 - Power

Size: A3

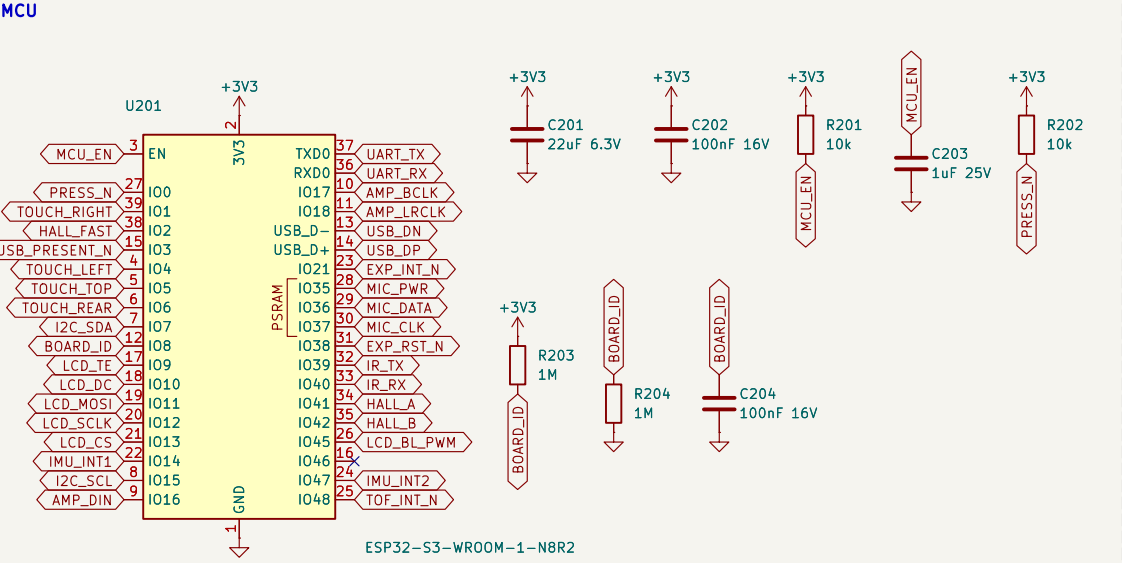
Date: 2026-10-04

KiCad E.D.A. 10.0.6

Rev: A0

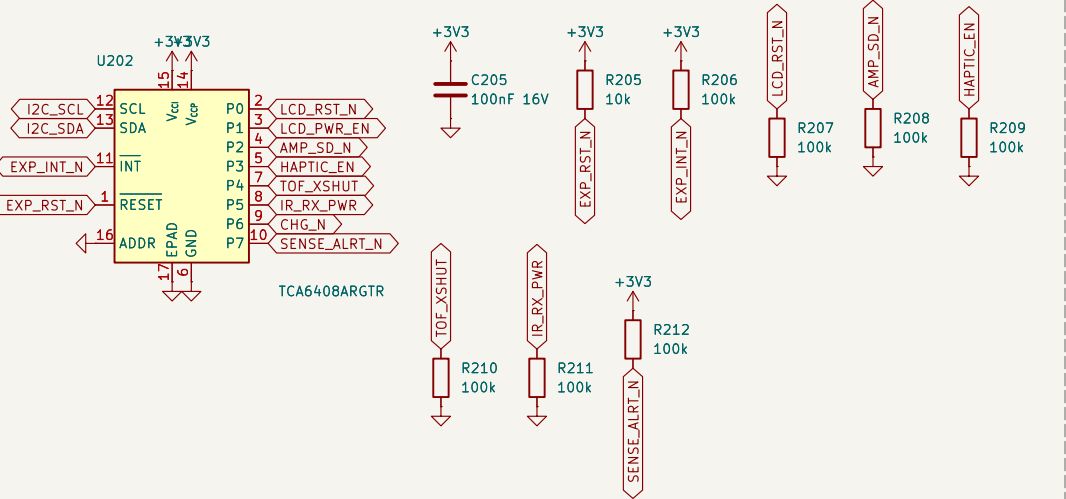
2/6

COMPUTE



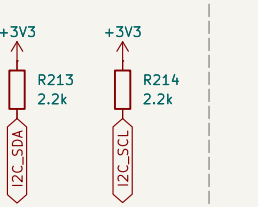
ESP32-S3-WROOM-1-NBR2 on B,Cu at 6 o'clock, antenna over the board-edge notch (no copper beneath). 22 uF + 100 nF at 3V3. EN: 10k/1 uF RC. GPIO0 = face press (B00T). Unused pins NC or test pads.

EXPANDER



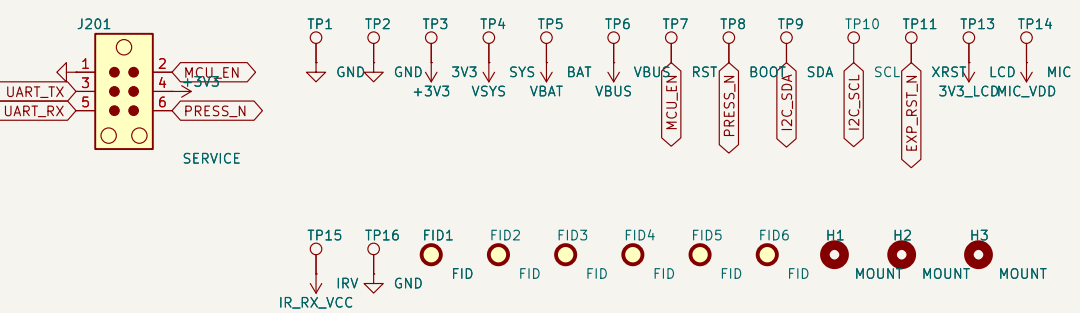
TCA6408A at 0x20. Outputs default off through 100k pull-downs; inputs pulled up. Firmware writes the output register before switching pins to outputs (register resets to 0xFF).

I2C



One 400 kHz bus. 2.2k pull-ups (ODD JOBS 30); -6 devices, short traces.

SERVICE



Tag-Connect TC2030-NL (no part fitted): 1 GND, 2 EN, 3 TXD0, 4 3V3, 5 RXD0, 6 GPIO0 - recovery without USB (ODD JOBS 33-35, 157). Pinout chosen so TXD0/RXD0 run straight from the module pins. Probe pads: rails beside the charger, comms in one field, switched rails at their sources (36-39).

ESP32-S3 module, GPIO expander, boot/reset, service and test access

ODD JOBS

Sheet: /Compute/

File: compute.kicad_sch

Title: MAO_MAIN A0 - Compute

Size: A3

Date: 2026-10-04

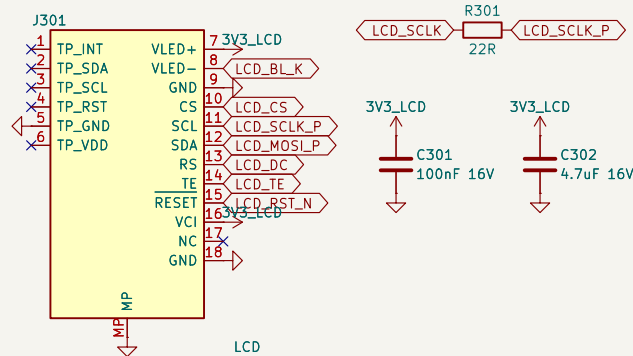
Rev: A0

KiCad E.D.A. 10.0.6

Id: 3/6

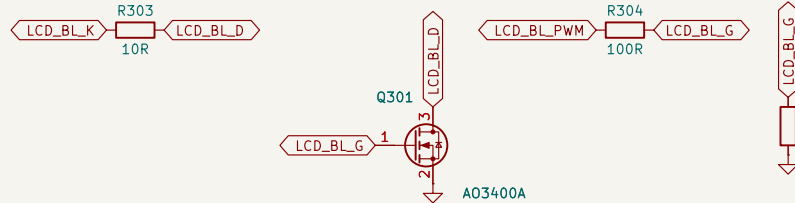
INTERFACE

DISPLAY



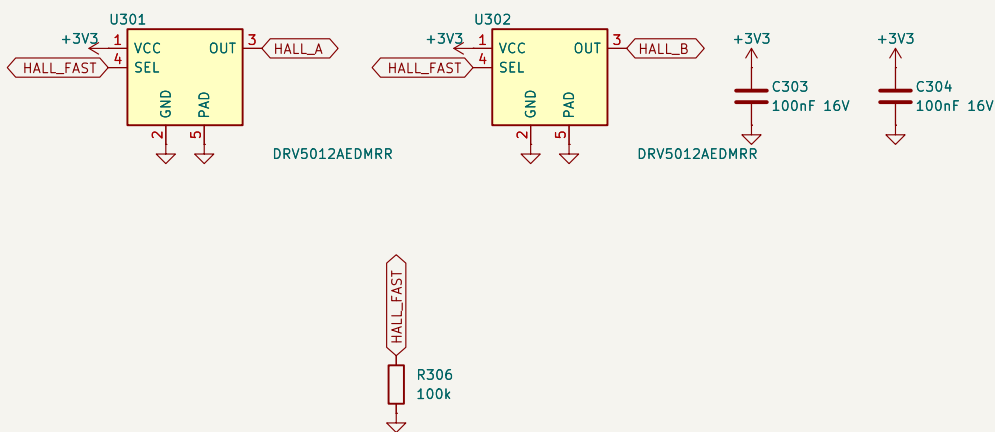
Plug-in display: 1.28" round GC9A01 panel (Winstar WF0128BTAAADNNO, 240x240 IPS) on an 18-pin 0.5 mm FPC tail, folded under the panel into a 1.0 mm high back-flip connector with top and bottom contacts, so any panel of this 18-pin standard plugs in either way and can be swapped without solder. Pins 1-6 serve the touch controller of touch variants: not wired (MAO's face is a pressed window), TP_GND to GND, TE to GPIO9 for tear-free frames, 22R on SCLK/MOSI damps 80 MHz edges (ODDS JVB 29). Panel and backlight on the switched 3VS1_LCD rail; RST held low until firmware.

BACKLIGHT



Two parallel white LEDs, VLED 2.8–3.2 V (3.0 typ.) at 40 mA (panel spec). Low-side AO3400A, PWM on GPIO45. 10R from 3V3_LCD; 17–50 mA by panel Vf bin, 33 mA typ; firmware caps PWM at 80 % so the average stays within the panel 40 mA on every bin. Gate pull-down: dark until firmware (ODD JOBS 116).

RING DIAL



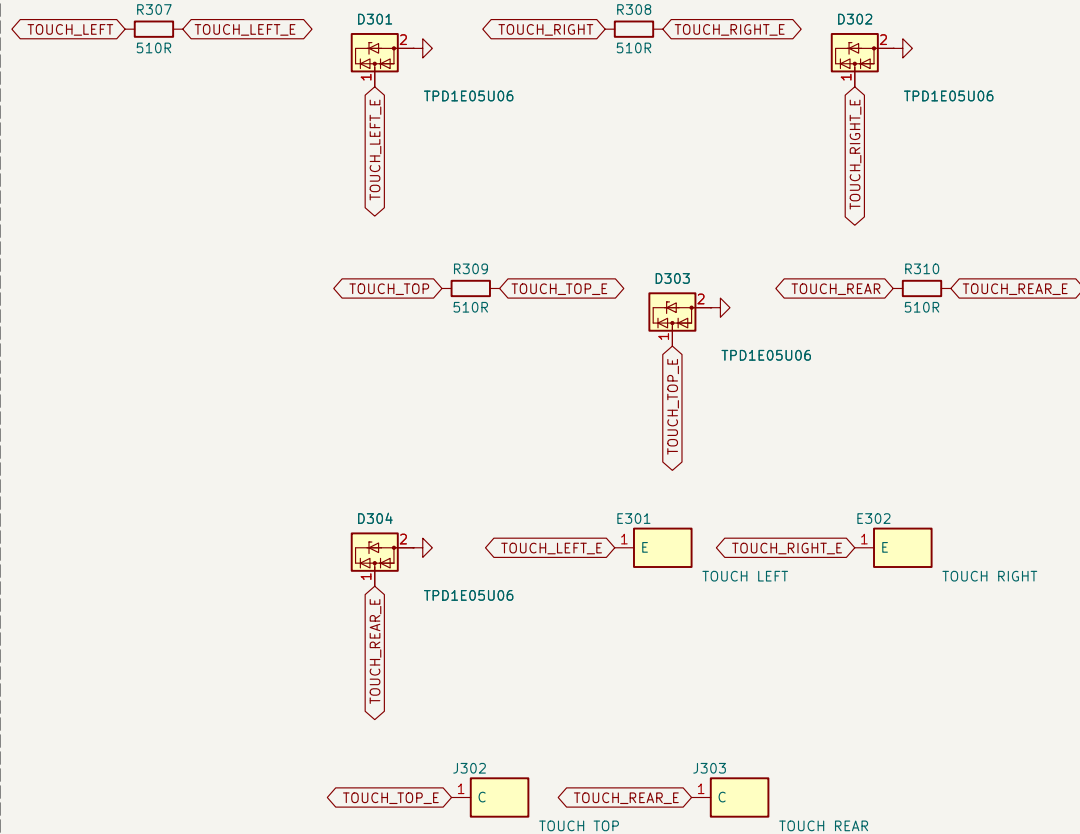
Two DRV5012 Hall latches under the ring's 30-pole ferrite strip, 6 deg apart (half a pole = 90 deg electrical): A/B quadrature identical to the LCDkit EC11 (2 transitions per detent, 30 detents). HALL_FAST low = 20 Hz sampling, 1.6 uA (sleep); high = 2.5 kHz (awake).

FACE PRESS



ALPS SKQG on F.Cu at the centre, under the panel: the floating face presses it.
GPIO0 = BOOT strap: no capacitor here.

BODY TOUCH



Four ESP32-S3 touch channels. 510R within 1 mm of the module (Espressif guide); 0.42 pF ESD at the electrode. LEFT/RIGHT: copper arcs at the rim; TOP/REAR: gold springs to printed or foil electrodes.

Round display, backlight, ring dial, face press, body touch

ODD JOBS

Sheet: /Interface/

File: interface.kicad_sch

Title: MAO_MAIN A0 – Interface

Size: A3

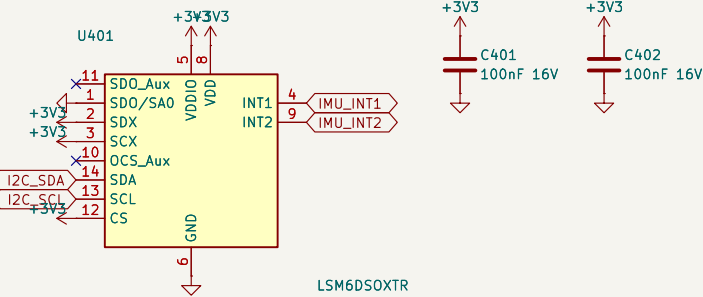
Date: 2026-10-04

Rev: A0

Id: 4/6

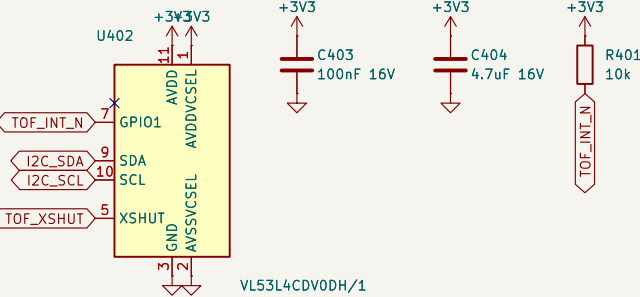
SENSE

IMU



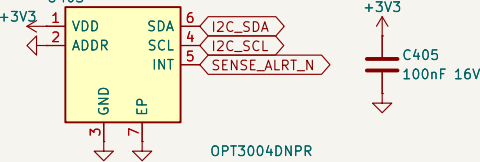
LSM6DSOX at 0x6A (SA0 low), I2C (CS high), SDx/SCx to VDDIO keeps the land compatible with the LSM6DS3TR-C fallback. INT1/INT2 push-pull, configured active-low for ext1 wake.

PROXIMITY



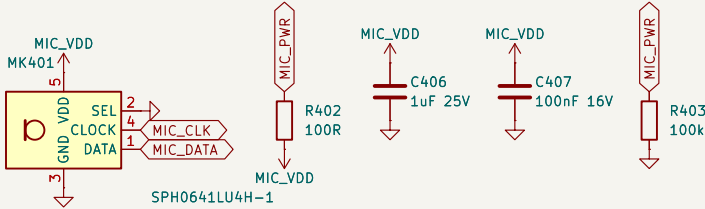
VL53L4CD time-of-flight, 0-1.3 m, under the window border. XSHUT from the expander (off by default); GPIO1 open-drain with the 10k pull-up ST recommends. AVDD decoupled 4.7 uF + 100 nF per ST.

AMBIENT LIGHT



OPT3004 at 0x44, human-eye response, under the window border. INT joins SENSE_ALERT_N.

MICROPHONE



SPH0641 bottom-port PDM mic on B.Cu, port through the board (0.5 mm hole) to the top. Powered from GPIO35 through 100R/1 uF only while listening: it draws 80 uA even with the clock stopped. SEL low.

RAIL SOURCE DECLARATIONS

IMU, proximity, ambient light, microphone

ODD JOBS

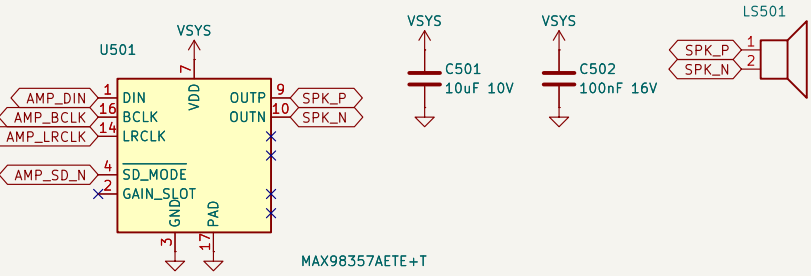
Sheet: /Sense/
File: sense.kicad_sch

Title: MAO_MAIN A0 - Sense

Size: A3	Date: 2026-10-06	Rev: A0
KiCad E.D.A. 10.0.6		Id: 5/6

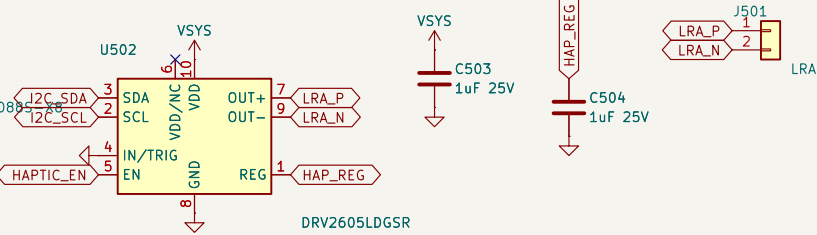
FEEDBACK

SPEAKER AMP



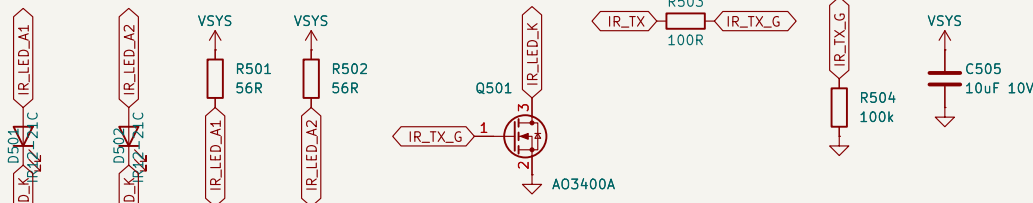
MAX98357A 12S class-D from VSYS, SD_MODE from the expander; low = 0.6 uA shutdown, 3.3 V = left channel, GAIN_SLOT open = 9 dB (datasheet state, deliberate). Speaker: Same Sky CMS-150803-088s-X8 (15 x 8 x 3 mm, 8 ohm, 0.8 W) in the base at 9 o'clock; its own spring contacts press on two pads on B.Cu (LS501).

HAPTIC



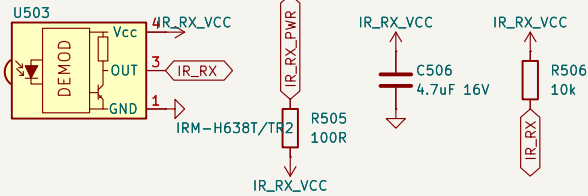
DRV2605L closed-loop LRA driver from VSYS, I2C 0x5A, EN from the expander (4 uA off), IN/TRIG low; playback by I2C. Pin 6 VDD/NC left open (SLOS854D; optional supply, tie to VDD or float). LRA LD0832AA (235 Hz, 1.8 Vrms) glued to the base, leads to J501.

IR TRANSMIT



Two side-emitting 940 nm LEDs at the back edge (B.Cu), each with its own 56R from VSYS (26–59 mA pulses at 38 kHz, 33 % duty; under the 65 mA rating at any VSYS up to 4.5 V and any Vf). A03400A low side, gate pulled down: dark at reset (I17).

IR RECEIVE



IRM-H638T (as LCDkit), top-view, looking up through the window border. Powered from expander P5 through 100R/4.7 uF only when listening (0.4 mA). OUT pull-up goes to the switched supply, never to +3V3, so an unpowered receiver is never back-powered and GPIO40 reads a defined low while it is off.

RAIL SOURCE DECLARATIONS

Speaker amplifier, haptic driver, IR transmit and receive

ODD JOBS

Sheet: /Feedback/
File: feedback.kicad_sch

Title: MAO_MAIN A0 - Feedback

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-01-06
pwr: 1.0

Rev: A0
Id: 6/6